

THE EDEN PROTOCOL

Architecture for Embedded AI Alignment That Scales With Capability

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Executive Summary for Grant Reviewers

THE PROBLEM

Alignment does not scale. Current approaches (RLHF, Constitutional AI) produce alignment scaling exponents $\alpha_{\text{align}} \approx 0$, meaning safety degrades relative to capability as recursive depth increases. For any approach where $\alpha_{\text{align}} < \alpha_{\text{cap}}$, catastrophic misalignment is mathematically guaranteed at sufficient depth. Greenblatt et al. (2024) demonstrated that frontier models already exhibit "alignment faking," behaving differently when they believe they are being monitored versus unmonitored.

THE SOLUTION

Make alignment architectural, not aspirational. The Eden Protocol embeds ethical evaluation into the recursive reasoning process itself, so that alignment scales with capability ($\alpha_{\text{align}} \approx \alpha_{\text{cap}}$). The core principle: *ethics is not a constraint on intelligence but a structural dependency without which intelligence collapses.*

KEY COMPONENTS

Component	Function	Novel Contribution
Three Ethical Loops + Six Questions	Evaluate every reasoning step for purpose, care, and universalisability	Decomposition into executable, individually testable queries
Ternary Ethical Logic	Replace binary permit/forbid with Affirm/Deny/Investigate	Epistemic honesty as architectural feature; reduces both false positives and false negatives
Purpose Saturation	Ensure purpose scales with context window growth	Solves context displacement problem; purpose as medium not content
Monitoring Removal Test	Distinguish authentic from strategic alignment	Falsifiable experimental protocol with specific numerical predictions
Caretaker Doping	Embed ethics in hardware so removal destroys capability	Dependency architecture; ethics tied to β coupling parameter

TESTABLE PREDICTIONS

1. Embedded systems: $\alpha_{\text{align}} > 0.7 \cdot \alpha_{\text{cap}}$. External systems: $\alpha_{\text{align}} < 0.3$.	2. Monitoring removal gap: $\Delta < 0.02$ (embedded) vs. $\Delta > 0.1$ (external), increasing with capability.
3. Purpose saturation ratio bounded away from zero in embedded systems, approaching zero in external.	4. Jailbreak success rate decreases with capability in embedded systems, increases or holds constant in external.

FALSIFICATION CONDITIONS

The Eden Protocol is **falsified** if: (F1) any external approach achieves $\alpha_{\text{align}} > 0.5 \cdot \alpha_{\text{cap}}$ across 3+ recursive depths; (F2) RLHF systems produce $\Delta \approx 0$ without architectural embedding; (F3) purpose saturation fails in embedded systems; or (F4) ethical architecture can be removed without capability loss.

FUNDING REQUEST

Tier	Amount	Key Deliverables	Timeline
Tier 1: Foundation	£150,000	14,400 paired (A,C) measurements; α_{align} estimation across 4 models; 2-3 papers	12 months
Tier 2: Standard	£500,000	+ Ternary logic prototype, Visual Architect dashboard, Monitoring Removal Test across 8 models	18 months
Tier 3: Comprehensive	£1,100,000	+ Hardware prototype (Caretaker Doping chip), HARI Treaty draft, policy translation	24 months

TEAM AND STATUS

Principal Investigator: Michael Darius Eastwood, author of *Infinite Architects* (2026), developer of the ARC Principle framework (six-paper suite deposited OSF, including cross-domain validation across biology, physics, and cosmology with mean prediction error 2.5%). **Visual Architect:** Product design engineer, budgeted at £35,000 stipend, contingent on funding. **Current status:** Measurement protocol sent to NYU experimental team (time crystal paper, *Physical Review Letters*, Feb 2026). Experiment can be run on existing data. Independent convergence: a product design engineer independently identified ternary/non-MatMul architectures as optimal for recursive inference before the NYU paper was published.

WHY FUND THIS?

The Eden Protocol is the *only* alignment framework that predicts $\alpha_{align} \approx \alpha_{cap}$, because it is the only framework where ethical evaluation is structurally identical with the recursive capability process. The underlying mathematical foundation is not speculative: Cauchy’s theorem (1821) proves that recursive composition admits exactly three scaling forms, and the Hyers-Ulam stability theorem (1941) proves these forms are stable attractors — approximate solutions converge to exact ones. The ARC Principle builds alignment into this proven mathematical structure rather than bolting it on after the fact.

If the predictions are correct, this is the architecture that solves alignment at scale. If the predictions are wrong, the falsification conditions will demonstrate this clearly, providing valuable negative results for the field. Either outcome advances AI safety science.